



Vergabepilot.AI

Project Leads

Collaborative Initiative (CORE & CICONIA)



PROF. DR. CHRISTIAN BARTELT



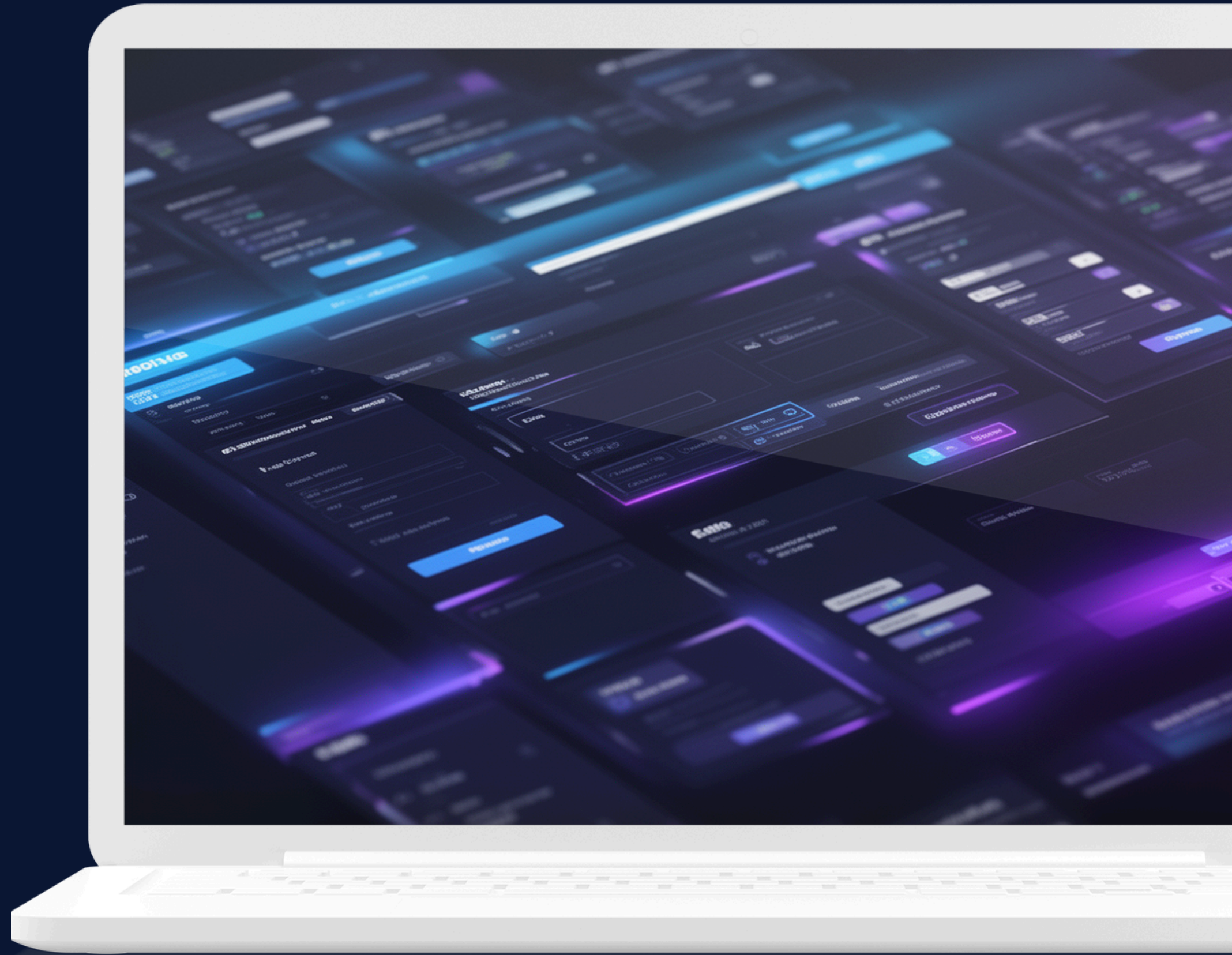
KRISTIAN KOLTHOFF



PROBLEM OVERVIEW

Problem and Motivation

Every new data source needs a scraper built from scratch. That is slow/costly and hard to scale. As domains grow, engineering effort becomes the bottleneck.



Romania Week	In Between Week	Goslar Week
• Kick-off (Team Meeting) • Hackathon ◦ Manual Scraping ◦ LLM-Assisted Scraping ◦ CUA Scraper • Model Comparsion	• Team Division • Annotated Dataset • Different pipeline Approaches	• Cascade Style Pipeline • Phase 1 ◦ Generator ◦ Sandbox (Executor) ◦ Validator ◦ Evaluator (beta) • Phase 2 ◦ Cua Scraper • Phase 3 ◦ Cascade-Style Pipeline

Pipeline Workflow

Phase 3

- Modular Approach
- Sequential Model

Phase 1

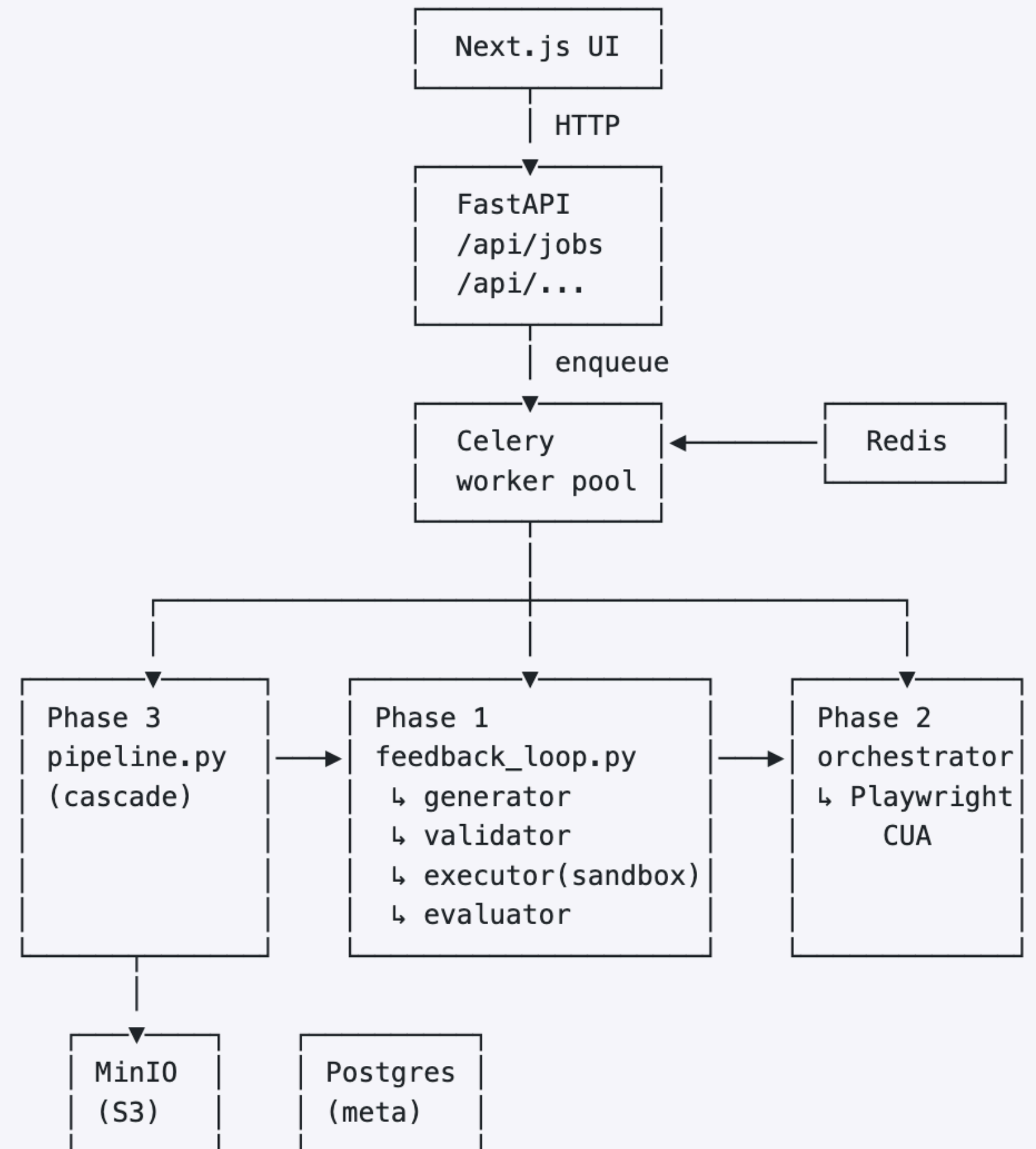
- Generates Scraper
- Feedback loop

Phase 2

- Visual CUA approach
- Fallback Option

Storage

- Persistent Document Storage (S3)



Phase-1: Scraper Feedback loop

Generator

- Takes a URL and generates a Python scraper via LLM per domain
- Describes the input or source system

Validator

- Checks code safety using AST analysis before execution
- Describes the processing or orchestration layer

Executor

- Runs scrapers in a sandboxed subprocess with memory and timeout limits
- Describes the output or delivery layer

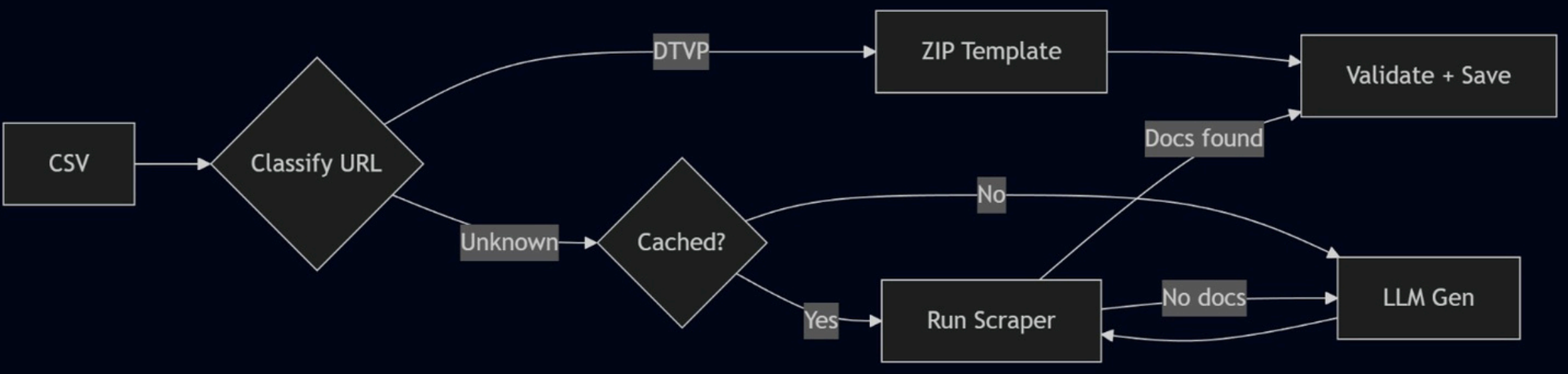
Evaluator

- Compares results against expected outcomes and flags discrepancies
- Describes any supporting infrastructure

Scraper Generation Pipeline

- URL Classification Pipeline
- Cache-Based Optimization
- Automated Scraper Execution
- LLM Fallback Generation
- Validation & Data Storage

Run	Cache	Success	LLM Calls	Cost	Time
Production <code>llm_cache/</code>	Iteratively generated	97/100 (97%)	4 retries	\$0.02	348s
Eval cache (fresh regen)	Single-shot generation	87/100 (87%)	79 fresh + 18 retries	\$0.53	1340s



```
1 {
2   "total": 100,
3   "success": 97,
4   "success_rate_pct": 97.0,
5   "elapsed_total_s": 348.4,
6   "by_status": {
7     "success": 97,
8     "no_documents": 3
9   },
10  "by_platform": {
11    "dtvp": 21,
12    "Unknown (PDF)": 2,
13    "NetServer": 35,
14    "evergabe-online.de": 1,
15    "vergabe24.de": 2,
16    "bi-medien.de": 2,
17    "aumass": 4,
18    "evergabe.de": 1,
19    "staatsanzeiger-eservices.de": 1,
20    "evergabe-online.de (Wicket)": 1,
21    "deutsche-evergabe.de": 3,
22    "meinauftrag.rib.de": 3,
23    "eVergabe 4.9 / Cosinex": 3,
24    "had.de": 1,
25    "eVergabe 4.9 / Cosinex family": 2,
26    "meinauftrag.rib.de (via iframe)": 1,
27    "subreport.de": 2,
28    "it-vergabe.eu": 1,
29    "TYP03": 3,
30    "Lubey": 1,
31    "Verwaltungsportal": 1,
32    "trading.energie-handelsplatz.de": 1,
33    "TYP03 (GSB11)": 1,
34    "e-VA Bieterportal (PD-G)": 1,
35    "SharePoint": 1,
36    "Unknown (ZIP file)": 1,
37    "laenderfinanzierungsprogramm.de": 1,
38    "Drupal": 1,
39    "Ariba": 1,
40    "acingov.pt": 1
41  },
42  "llm_new_calls": 4,
43  "llm_cached_calls": 75
44 }
```

Multi-Model Benchmarking

- Best Model: Gemini Flash Lite
- 83% Extraction Success
- Lowest Cost, Fastest Speed
- Expensive Models Underperformed

Rank	Model	Success%	Latency	Cost/run	Grade
#1	Gemini 2.5 Flash Lite	83%	4,176 ms	\$0.00181	A
#2	Gemini 2.5 Flash	83%	4,540 ms	\$0.00338	A
#3	GPT-4o mini	67%	12,786 ms	\$0.00273	B
#4	Claude Sonnet	67%	17,705 ms	\$0.07191	B
#5	Claude Haiku	67%	7,782 ms	\$0.01867	B
#6	GPT-4.1	67%	15,329 ms	\$0.03993	B
#7	GPT-4.1 mini	67%	19,224 ms	\$0.00798	B-
#8	DeepSeek V3	67%	46,581 ms	\$0.00238	C
#9	Gemini 2.5 Pro	50%	24,989 ms	\$0.05495	D
#10	Gemini 3.1 Pro	0%	603 ms	\$0.00000	F

Demo

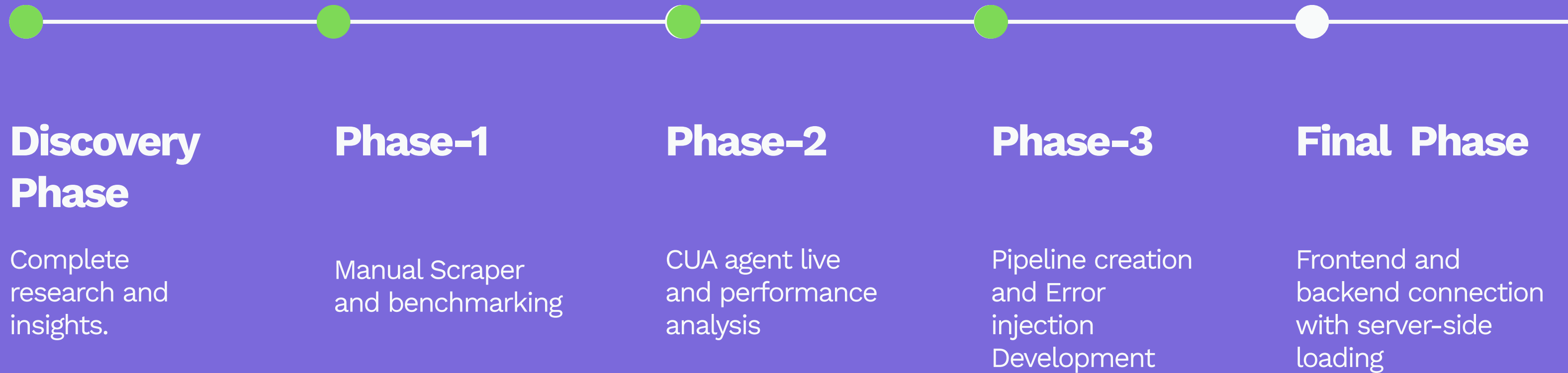
```
Loaded 100 URLs
```

```
Processing 100 URLs with 4 workers ...
```

```
Output: C:\vergabepilot\Vergabepilot-v1\results\run_20260512_115859.json1
```

```
[ 1/100] OK success      docs=1  platform=NetServer      https://www.ausschreibungen.ls.brandenburg.de/NetServer/PublicationCon
```

Project Status & Roadmap





Our Talented Team

Thank You

This presentation highlighted key phases of Vergabepilot.AI, emphasizing collaboration and innovative AI solutions for project scraping.



Vergabepilot.AI